The effects of noble gas intercalation on bilayer MoS_2 sheets: A DFT study

P. James[1] E. H. Åhlgren[1]

[1]University of Helsinki

2D MoS_2 is a material of significant interest because its electronic band structure and catalytic activity can be tuned by modifying layer thickness or introducing structural defects such as strain and vacancies in the lattice. This opens interesting possibilities for creating new MoS_2-systems with targeted properties. This work investigates the effects of noble gas (He, Ar, Xe) intercalation on bilayer MoS_2 using ab initio density functional theory (DFT) simulations with the Quantum Espresso software package. We investigate bilayer supercells of varying dimensions (1×1, 2×2, 3×3, and 4×4) each containing a single intercalated noble gas atom, resulting in systems with different noble gas densities between the layers. The electronic band gap structure is resolved for each system and compared against pristine bilayer (indirect bandgap of ∼1.6eV) and monolayer MoS_2 structures (direct bandgap of ∼1.9 eV) [1]. The results indicate that noble gas intercalation causes the bandgap of bilayer MoS_2 to approach that of pristine monolayer MoS_2 as supercell size increases and noble gas atomic density decreases. Furthermore, the bandgap in all intercalated systems is indirect, consistent with the behavior observed in all pristine MoS_2 structures thicker than a monolayer. There is a notable exception with the 3×3×2 systems, in which noble gas intercalation produces a direct bandgap regardless of the noble gas species.

[1] Mak et al, *APS* **105**, 6 (2010).